

Revision date: 23 Jun 2025 Version: 2 Print date: 12 Mar 2026

Safety Data Sheet according to Regulation (EC) No. 1907/2006 (REACH)

SECTION 1: Identification of the substance/mixture and of the company/undertaking

1.1. Product identifier

Trade name/designation:

Cyclus® Bead Extraction, Wash Buffer 1

Article No.:

SMB95-6000

UFI:

AFRU-72WY-Q6SE-PT09

1.2. Relevant identified uses of the substance or mixture and uses advised against

Use of the substance/mixture:

The product is intended for research, analysis and scientific education.

1.3. Details of the supplier of the safety data sheet

Supplier (manufacturer/importer/only representative/downstream user/distributor):

Minerva Biolabs GmbH

Schkopauer Ring 13

12681 Berlin

Germany

Telephone: +49 30 2000437 0

E-mail: ehs@minerva-biolabs.com

Website: www.minerva-biolabs.com

1.4. Emergency telephone number

Vergiftungs-Informations-Zentrale Freiburg, 24h: +49 761 19240

SECTION 2: Hazards identification

2.1. Classification of the substance or mixture

Classification according to Regulation (EC) No 1272/2008 [CLP]

Hazard classes and hazard categories	Hazard statements	Classification procedure
Acute toxicity (oral) (<i>Acute Tox. 4</i>)	H302: Harmful if swallowed.	Calculation method.
Skin corrosion/irritation (<i>Skin Irrit. 2</i>)	H315: Causes skin irritation.	Calculation method.
Serious eye damage/eye irritation (<i>Eye Irrit. 2</i>)	H319: Causes serious eye irritation.	Calculation method.
Acute toxicity (inhalative) (<i>Acute Tox. 4</i>)	H332: Harmful if inhaled.	Calculation method.

* 2.2. Label elements

Labelling according to Regulation (EC) No. 1272/2008 [CLP]

Hazard pictograms:



GHS07

Exclamation mark

Signal word: Warning

Hazard statements for health hazards	
H302 + H332	Harmful if swallowed or if inhaled.
H315	Causes skin irritation.
H319	Causes serious eye irritation.
H332	Harmful if inhaled.

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Supplemental hazard information: none

Precautionary statements Prevention

P261	Avoid breathing dust/fume/gas/mist/vapours/spray.
P264	Wash hands thoroughly after handling.
P270	Do not eat, drink or smoke when using this product.
P280	Wear protective gloves/protective clothing and eye protection/face protection.

Precautionary statements Response

P301 + P312	IF SWALLOWED: Call a POISON CENTER/doctor/First aider if you feel unwell.
P302 + P352	IF ON SKIN: Wash with plenty of water/soap.
P304 + P340	IF INHALED: Remove person to fresh air and keep comfortable for breathing.
P305 + P351 + P338	IF IN EYES: Rinse cautiously with water for several minutes. Remove contact lenses, if present and easy to do. Continue rinsing.
P312	Call a POISON CENTER/doctor/First aider if you feel unwell.
P332 + P313	If skin irritation occurs: Get medical advice/attention.
P337 + P313	If eye irritation persists: Get medical advice/attention.

Precautionary statements Disposal

P501	Dispose of contents/container to an appropriate recycling or disposal facility.
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2.3. Other hazards

No data available

SECTION 3: Composition/information on ingredients

3.2. Mixtures

Hazardous ingredients / Hazardous impurities / Stabilisers:

Product identifiers	Substance name Classification according to Regulation (EC) No 1272/2008 [CLP]	Concentration
CAS No.: 50-01-1 EC No.: 200-002-3 Index No.: 607-148-00-0	guanidinium chloride Acute Tox. 4 (H302), Eye Irrit. 2 (H319), Skin Irrit. 2 (H315)  Warning Acute Toxicity Estimate ATE (oral) 729.8 mg/kg ATE (dermal) > 2,000 mg/kg ATE (inhalation, dust/mist) 3.181 mg/L	10 – ≤ 25 weight-%
CAS No.: 137-16-6 EC No.: 205-281-5 REACH No.: 01-2119527780-39-XXXX	sodium N-lauroylsarcosinate Acute Tox. 2 (H330), Eye Dam. 1 (H318), Eye Irrit. 2 (H319), Skin Irrit. 2 (H315)   Danger Specific concentration limit (SCL) Skin Irrit. 2; H315: C ≥ 30% Eye Dam. 1; H318: C ≥ 30% Eye Irrit. 2; H319: 1% ≤ C < 30% Acute Toxicity Estimate ATE (oral) > 5,000 mg/kg ATE (inhalation, dust/mist) > 0.05 – < 0.5 mg/L	> 2.5 – < 4.5 weight-%

Full text of H- and EUH-phrases: see section 16.

SECTION 4: First aid measures

4.1. Description of first aid measures

General information:

In case of accident or unwellness, seek medical advice immediately (show directions for use or safety data sheet if possible). Remove victim out of the danger area. Remove contaminated, saturated clothing immediately. Do not leave affected person unattended. If unconscious but breathing normally, place in recovery position and seek medical advice.

Following inhalation:

Provide fresh air. In case of respiratory tract irritation, consult a physician.

In case of skin contact:

After contact with skin, wash immediately with plenty of water and soap. Take off immediately all contaminated clothing. If skin irritation or rash occurs: Get medical advice/attention.

After eye contact:

Rinse cautiously with water for several minutes. Remove contact lenses, if present and easy to do. Continue rinsing. Call a physician immediately.

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Following ingestion:

Rinse mouth. Let 1 glass of water be drunken in little sips (dilution effect). Call a physician immediately.

Self-protection of the first aider:

Use personal protection equipment. No direct artificial respiration to be given by first aider.

4.2. Most important symptoms and effects, both acute and delayed

Serious eye damage/eye irritation, Skin corrosion/irritation

4.3. Indication of any immediate medical attention and special treatment needed

Treat symptomatically.

SECTION 5: Firefighting measures

5.1. Extinguishing media

Suitable extinguishing media:

Water spray jet, Carbon dioxide (CO₂), alcohol resistant foam

Unsuitable extinguishing media:

Full water jet

5.2. Special hazards arising from the substance or mixture

Hazardous combustion products:

In case of fire: Gases/vapours, toxic

* **5.3. Advice for firefighters**

Use suitable breathing apparatus. Fight fire with normal precautions from a reasonable distance. Do not inhale explosion and combustion gases.

5.4. Additional information

Collect contaminated fire extinguishing water separately. Do not allow entering drains or surface water.

SECTION 6: Accidental release measures

* **6.1. Personal precautions, protective equipment and emergency procedures**

6.1.1. For non-emergency personnel

Personal precautions:

Remove persons to safety. Use personal protective equipment as required. Avoid contact with skin, eyes and clothes. Provide adequate ventilation.

Protective equipment:

Wear protective gloves/protective clothing and eye protection/face protection.

6.1.2. For emergency responders

Personal protection equipment:

Personal protection equipment: see section 8

6.2. Environmental precautions

Do not allow to enter into surface water or drains.

* **6.3. Methods and material for containment and cleaning up**

For containment:

Provide adequate ventilation. Absorb with liquid-binding material (sand, diatomaceous earth, acid- or universal binding agents).

For cleaning up:

Water (with cleaning agent)

6.4. Reference to other sections

Safe handling: see section 7, Personal protection equipment: see section 8, Disposal: see section 13

6.5. Additional information

Use appropriate container to avoid environmental contamination.

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SECTION 7: Handling and storage

* 7.1. Precautions for safe handling

Protective measures

Advices on safe handling:

Wear personal protection equipment (refer to section 8). Wash hands before breaks and after work. Keep away from: Food and feedingstuffs. If local exhaust ventilation is not possible or not sufficient, the entire working area should be ventilated by technical means. Always close containers tightly after the removal of product.

Advices on general occupational hygiene

Avoid contact with skin, eyes and clothes. When using do not eat, drink, smoke, sniff.

7.2. Conditions for safe storage, including any incompatibilities

Technical measures and storage conditions:

Keep container tightly closed in a cool, well-ventilated place.

Requirements for storage rooms and vessels:

Keep container tightly closed in a cool, well-ventilated place. Recommended storage temperature: at room temperature. Protect from sunlight.

Hints on storage assembly:

Do not store together with: Acid, Food and feedingstuffs

Storage class (TRGS 510, Germany): 12 – non-combustible liquids that cannot be assigned to any of the above storage classes

Further information on storage conditions:

Keep container tightly closed in a cool, well-ventilated place.

7.3. Specific end use(s)

No data available

SECTION 8: Exposure controls/personal protection

8.1. Control parameters

No data available

* 8.2. Exposure controls

8.2.1. Appropriate engineering controls

If local exhaust ventilation is not possible or not sufficient, the entire working area should be ventilated by technical means.

8.2.2. Personal protection equipment



Eye/face protection:

Eye glasses with side protection (EN 166).

Skin protection:

Tested protective gloves must be worn (EN ISO 374). Take recovery periods for skin regeneration. Breakthrough times and swelling properties of the material must be taken into consideration. Suitable material: NBR (Nitrile rubber), Thickness of the glove material: > 0,35 mm. In the case of wanting to use the gloves again, clean them before taking off and air them well.

Respiratory protection:

Usually no personal respiratory protection necessary. In case of inadequate ventilation wear respiratory protection. Filter type: ABEK-P3

Other protection measures:

Wash hands before breaks and after work.

8.2.3. Environmental exposure controls

No data available

SECTION 9: Physical and chemical properties

* 9.1. Information on basic physical and chemical properties

Appearance

Physical state: Liquid

Colour: colourless

Odour: not determined

flammability: No

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Safety relevant basis data

Parameter	Value	at °C	① Method ② Remark
pH	6		
Melting point	No data available		
Freezing point	No data available		
Initial boiling point and boiling range	100 °C		
Flash point	No data available		
Evaporation rate	No data available		
Auto-ignition temperature	No data available		
Upper/lower flammability or explosive limits	No data available		
Vapour pressure	23 hPa	20 °C	
Vapour density	No data available		
Density	1.044 g/cm³	20 °C	
Bulk density	not applicable		
Water solubility	completely miscible		
Dynamic viscosity	No data available		
Kinematic viscosity	No data available		

9.2. Other information

No data available

SECTION 10: Stability and reactivity

10.1. Reactivity

No specific test data related to reactivity available for this product or its ingredients.

10.2. Chemical stability

The product is chemically stable under recommended conditions of storage, use and temperature.

10.3. Possibility of hazardous reactions

Contact with acids liberates toxic gas.

10.4. Conditions to avoid

Further information on proper storage: see section 7.

10.5. Incompatible materials

No further relevant information available.

10.6. Hazardous decomposition products

Carbon dioxide (CO₂), Carbon monoxide

SECTION 11: Toxicological information

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11.1. Information on hazard classes as defined in Regulation (EC) No 1272/2008

Toxicological information

Mixture related information
LD ₅₀ oral: 3,041mg/kg (Rat)
LC ₅₀ Acute inhalation toxicity (vapour): 6.4mg/L 4h
guanidinium chloride CAS No.: 50-01-1 EC No.: 200-002-3
LD ₅₀ oral: 729.8mg/kg (rat) other: Health effects test guidelines, August 1982, EPA 560/6-82-001
LD ₅₀ dermal: >2,000mg/kg (rabbit) other: EPA TS-792 Acute exposure, dermal toxicity. Health effects test guidelines, EPA, August 1982; EPA 560/6-82-001
LC ₅₀ Acute inhalation toxicity (dust/mist): 3.181mg/L 4h (Rat)
sodium N-lauroylsarcosinate CAS No.: 137-16-6 EC No.: 205-281-5
LD ₅₀ oral: >5,000mg/kg (rat) OECD Guideline 401 (Acute Oral Toxicity)
LC ₅₀ Acute inhalation toxicity (dust/mist): >0.05 – <0.5mg/L 4h (rat) OECD Guideline 403 (Acute Inhalation Toxicity)

Acute oral toxicity:

Harmful if swallowed.

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Acute dermal toxicity:

Based on available data, the classification criteria are not met.

Acute inhalation toxicity:

Harmful if inhaled.

Skin corrosion/irritation:

Causes skin irritation.

Serious eye damage/irritation:

Causes serious eye irritation.

Respiratory or skin sensitisation:

Based on available data, the classification criteria are not met.

Germ cell mutagenicity:

Based on available data, the classification criteria are not met.

Carcinogenicity:

Based on available data, the classification criteria are not met.

Reproductive toxicity:

Based on available data, the classification criteria are not met.

STOT-single exposure:

Based on available data, the classification criteria are not met.

STOT-repeated exposure:

Based on available data, the classification criteria are not met.

Aspiration hazard:

Based on available data, the classification criteria are not met.

Additional information:

No data available

* **11.2. Information on other hazards**

Endocrine disrupting properties:

This product does not contain a substance that has endocrine disrupting properties with respect to humans as no components meets the criteria.

SECTION 12: Ecological information

* **12.1. Toxicity**

guanidinium chloride CAS No.: 50-01-1 EC No.: 200-002-3
LC₅₀: 2,420mg/L 2d (fish, <i>Leuciscus idus</i>) other: DIN 38412 L15 (DEV-11. Lieferung 1982)
EC₅₀: 33.5mg/L 3d (Algae/water plant, <i>Raphidocelis subcapitata</i> (previous names: <i>Pseudokirchneriella subcapitata</i> , <i>Selenastrum capricornutum</i>)) EU Method C.3 (Algal Inhibition test)
EC₅₀: 70.2mg/L 2d (crustaceans, <i>Daphnia magna</i>) OECD Guideline 202 (<i>Daphnia</i> sp. Acute Immobilisation Test)
NOEC: 6.3mg/L 3d (Algae/water plant, <i>Raphidocelis subcapitata</i> (previous names: <i>Pseudokirchneriella subcapitata</i> , <i>Selenastrum capricornutum</i>)) EU Method C.3 (Algal Inhibition test)
NOEC: ≥181mg/L 35d (fish, <i>Pimephales promelas</i>) OECD Guideline 210 (Fish, Early-Life Stage Toxicity Test)
NOEC: 2.9mg/L 21d (crustaceans, <i>Daphnia magna</i>) OECD Guideline 211 (<i>Daphnia magna</i> Reproduction Test)
LOEC: ≥424mg/L 35d (fish, <i>Pimephales promelas</i>) OECD Guideline 210 (Fish, Early-Life Stage Toxicity Test)
LOEC: 4.2mg/L 21d (crustaceans, <i>Daphnia magna</i>) OECD Guideline 211 (<i>Daphnia magna</i> Reproduction Test)
sodium N-lauroylsarcosinate CAS No.: 137-16-6 EC No.: 205-281-5
LC₅₀: 32.1mg/L 4d (fish, <i>Danio rerio</i> (previous name: <i>Brachydanio rerio</i>)) OECD Guideline 203 (Fish, Acute Toxicity Test)
EC₅₀: 39mg/L 3d (Algae/water plant, <i>Desmodesmus subspicatus</i> (previous name: <i>Scenedesmus subspicatus</i>)) OECD Guideline 201 (Alga, Growth Inhibition Test)
EC₅₀: 8.91mg/L 2d (crustaceans, <i>Daphnia magna</i>) OECD Guideline 202 (<i>Daphnia</i> sp. Acute Immobilisation Test)
NOEC: 9.2mg/L 3d (Algae/water plant, <i>Desmodesmus subspicatus</i> (previous name: <i>Scenedesmus subspicatus</i>)) OECD Guideline 201 (Alga, Growth Inhibition Test)
NOEC: 50mg/L 4d (fish, <i>Danio rerio</i> (previous name: <i>Brachydanio rerio</i>)) OECD Guideline 203 (Fish, Acute Toxicity Test)
NOEC: 5mg/L 2d (crustaceans, <i>Daphnia magna</i>) OECD Guideline 202 (<i>Daphnia</i> sp. Acute Immobilisation Test)
LOEC: 29mg/L 3d (Algae/water plant, <i>Desmodesmus subspicatus</i> (previous name: <i>Scenedesmus subspicatus</i>)) OECD Guideline 201 (Alga, Growth Inhibition Test)

12.2. Persistence and degradability

No data available

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12.3. Bioaccumulative potential

guanidinium chloride	CAS No.: 50-01-1	EC No.: 200-002-3
Log K _{OW} :	< 0.02	
Bioconcentration factor (BCF):	0.893	
sodium N-lauroylsarcosinate	CAS No.: 137-16-6	EC No.: 205-281-5
Log K _{OW} :	0.37	

12.4. Mobility in soil

No data available

12.5. Results of PBT and vPvB assessment

guanidinium chloride	CAS No.: 50-01-1	EC No.: 200-002-3
Results of PBT and vPvB assessment:	—	
sodium N-lauroylsarcosinate	CAS No.: 137-16-6	EC No.: 205-281-5
Results of PBT and vPvB assessment:	—	

12.6. Endocrine disrupting properties

No data available

12.7. Other adverse effects

No data available

SECTION 13: Disposal considerations

13.1. Waste treatment methods

Dispose of waste according to applicable legislation.

Waste treatment options

Appropriate disposal / Product:

Consult the appropriate local waste disposal expert about waste disposal.

Appropriate disposal / Package:

Completely emptied packages can be recycled.

SECTION 14: Transport information

Land transport (ADR/RID)	Inland waterway craft (ADN)	Sea transport (IMDG)	Air transport (ICAO-TI / IATA-DGR)
14.1. UN number or ID number			
No dangerous good in sense of these transport regulations.	No dangerous good in sense of these transport regulations.	No dangerous good in sense of these transport regulations.	No dangerous good in sense of these transport regulations.
14.2. UN proper shipping name			
No dangerous good in sense of these transport regulations.	No dangerous good in sense of these transport regulations.	No dangerous good in sense of these transport regulations.	No dangerous good in sense of these transport regulations.
14.3. Transport hazard class(es)			
not relevant	not relevant	not relevant	not relevant
14.4. Packing group			
not relevant	not relevant	not relevant	not relevant
14.5. Environmental hazards			
not relevant	not relevant	not relevant	not relevant
14.6. Special precautions for user			
not relevant	not relevant	not relevant	not relevant

14.7. Maritime transport in bulk according to IMO instruments

No data available

SECTION 15: Regulatory information

15.1. Safety, health and environmental regulations/legislation specific for the substance or mixture

15.1.1. EU legislation

Other regulations (EU):

Directive 2012/18/EU on the control of major-accident hazards involving dangerous substances [Seveso-III-Directive]:

This product is not assigned to a hazard category.

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1907/2006 REACH, 1272/2008 CLP GHS, 98/24/EC

15.1.2. National regulations

[DE] National regulations

Restrictions of occupation

22 JArbSchG., 4 MuSchRiV., 5 MuSchRiV., Betriebssicherheitsverordnung (BetrSichV)

Störfallverordnung (12. BlmschV)

for substances contained in the product:

This product is not assigned to a hazard category.

Water hazard class

WGK:

1 - slightly hazardous to water

Berufsgenossenschaftliche Vorschriften (DGUV-Vorschriften)

Berufsgenossenschaftliche Vorschriften (DGUV-Vorschriften)

Other regulations, restrictions and prohibition regulations

Hazardous Substances Ordinance (GefStoffV)

15.2. Chemical Safety Assessment

No data available

SECTION 16: Other information

16.1. Indication of changes

2.2.	Label elements
3.2.	Mixtures
4.1.	Description of first aid measures
5.3.	Advice for firefighters
6.1.	Personal precautions, protective equipment and emergency procedures
6.3.	Methods and material for containment and cleaning up
7.1.	Precautions for safe handling
8.2.	Exposure controls
9.1.	Information on basic physical and chemical properties
11.1.	Information on hazard classes as defined in Regulation (EC) No 1272/2008
11.2.	Information on other hazards
12.1.	Toxicity
14.8.	Additional information
15.3.	Additional information

16.2. Abbreviations and acronyms

ADN	European Agreement concerning the International Carriage of Dangerous Goods by Inland Waterways
ADR	European Agreement concerning the International Carriage of Dangerous Goods by Road
BCF	Bioconcentration Factor
CAS	Chemical Abstracts Service
CLP	Classification, Labelling and Packaging
DIN	German Institute for Standardization / German Industrial Standard
EC ₅₀	Effective Concentration 50%
ECHA	European Chemicals Agency
EN	European Standard
GHS	Globally Harmonized System of Classification and Labelling of Chemicals
ICAO	International Civil Aviation Organization
IMDG	International Maritime Dangerous Goods
IMO	International Maritime Organization
ISO	International Standards Organisation
LC ₅₀	Lethal (fatal) Concentration 50%
LD ₅₀	Lethal (fatal) Dose 50%
NFPA	National Fire Protection Association
NOEC	No Observed Effect Concentration
OECD	Organisation for Economic Cooperation and Development
PBT	persistent and bioaccumulative and toxic
REACH	Registration, Evaluation and Authorization of Chemicals
RID	Dangerous goods regulations for transport by rail
SCL	Specific concentration limit
TRGS	Technische Regeln für Gefahrstoffe

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UN United Nations

For abbreviations and acronyms, see: ECHA Guidance on information requirements and chemical safety assessment, chapter R.20 (Table of terms and abbreviations).

16.3. Key literature references and sources for data

TRGS 510, TRGS 525, TRGS 900, Safety data sheets for the ingredients, 1907/2006 REACH, 1272/2008 CLP GHS

Substance name	Type	source of supply
guanidinium chloride CAS No.: 50-01-1 EC No.: 200-002-3	LD ₅₀ oral; LD ₅₀ dermal; LC ₅₀ ; EC ₅₀ ; NOEC; LOEC	Source: European Chemicals Agency, http://echa.europa.eu/
sodium N-lauroylsarcosinate CAS No.: 137-16-6 EC No.: 205-281-5	LD ₅₀ oral; LC ₅₀ Acute inhalation toxicity (dust/mist); LC ₅₀ ; EC ₅₀ ; NOEC; LOEC	Source: European Chemicals Agency, http://echa.europa.eu/

16.4. Classification for mixtures and used evaluation method according to regulation (EC) No 1272/2008 [CLP]

Hazard classes and hazard categories	Hazard statements	Classification procedure
Acute toxicity (oral) (<i>Acute Tox. 4</i>)	H302: Harmful if swallowed.	Calculation method.
Skin corrosion/irritation (<i>Skin Irrit. 2</i>)	H315: Causes skin irritation.	Calculation method.
Serious eye damage/eye irritation (<i>Eye Irrit. 2</i>)	H319: Causes serious eye irritation.	Calculation method.
Acute toxicity (inhalative) (<i>Acute Tox. 4</i>)	H332: Harmful if inhaled.	Calculation method.

16.5. List of relevant hazard statements and/or precautionary statements from sections 2 to 15

Hazard statements	
H302	Harmful if swallowed.
H315	Causes skin irritation.
H318	Causes serious eye damage.
H319	Causes serious eye irritation.
H330	Fatal if inhaled.

16.6. Training advice

No data available

16.7. Additional information

This data sheet was prepared in accordance with EU Regulation (EC) No. 1907/2006 (REACH) and, where applicable, US Regulation 29 CFR 1910.1200, Appendix D.

The information is based on the level of knowledge and experience on the date of issue, it do not have the importance of assured properties. The information may not be changed or transferred to other products. Reproduction in the unchanged state is permitted.

* Data changed compared with the previous version.